

Database Design Document

SentinelCut — AI-Powered Audio/Video Profanity Censorship Platform

Document Version: 1.0 **Status:** Final for Implementation **Database Engine:** PostgreSQL 15+ (via Supabase) **Based On:** SRS v1.0, System Architecture v1.0

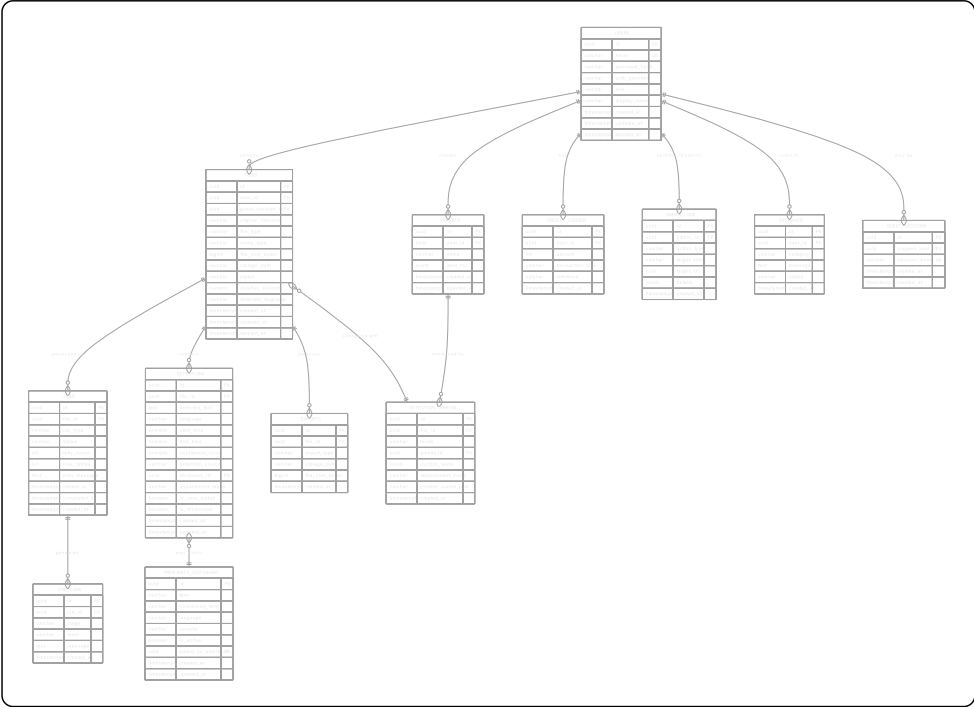
1. Purpose

This document defines the complete relational data model for SentinelCut: entity-relationship diagram, table schemas with column-level constraints, relationships, indexing strategy, and data integrity rules. It is the authoritative reference for implementing the Repository Layer defined in the System Architecture document.

2. Design Principles

- **Normalization to 3NF** for core relational entities (users, files, jobs) to avoid data anomalies, with **selective denormalization via JSONB** for flexible/semi-structured data (e.g., detection metadata snapshots, admin analytics caches) where strict normalization would add unnecessary join overhead.
- **UUID primary keys** throughout (not auto-increment integers) — required for Supabase Row-Level Security (RLS) compatibility and to avoid exposing sequential IDs (enumeration risk) in signed URLs/API responses.
- **Soft deletes** (`deleted_at` timestamp) on user-facing entities where audit history matters (files, users), rather than hard deletes, except where SRS-NFR-LEGAL-02 (right to delete account data) requires actual purging.
- **Every table has** `created_at` `updated_at` for auditability and debugging (supports SRS-NFR observability requirements).
- **Foreign keys with explicit** `ON DELETE` **behavior** defined per relationship (cascade vs. restrict) to prevent orphaned records.

3. Entity-Relationship Diagram



4. Table Schemas (DDL-Level Detail)

4.1 `users`

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
email	VARCHAR(255)	UNIQUE, NOT NULL
password_hash	VARCHAR(255)	NULL (nullable for OAuth-only accounts)
auth_provider	VARCHAR(20)	NOT NULL, CHECK IN ('email','google')
role	VARCHAR(20)	NOT NULL, DEFAULT 'user', CHECK IN ('user','admin')
display_name	VARCHAR(100)	NULL
storage_used_bytes	BIGINT	NOT NULL, DEFAULT 0
storage_limit_bytes	BIGINT	NOT NULL, DEFAULT 5368709120 (5GB default)
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()
deleted_at	TIMESTAMPTZ	NULL (soft delete)

Notes: `password_hash` is nullable because Google OAuth users authenticate without a local password (per FR-AUTH-02). Actual account-deletion requests (SRS-NFR-LEGAL-02) trigger a separate hard-delete routine that purges PII beyond just setting `deleted_at`.

4.2 `guest_sessions`

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
mapped_user_id	UUID	FK → users.id, NULL (guests don't have a permanent user row; may later map to one on signup)
session_token	VARCHAR(255)	UNIQUE, NOT NULL
expires_at	TIMESTAMPTZ	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

Notes: A scheduled job purges expired guest sessions and their associated `files`/`storage` records (satisfies FR-AUTH-07 and NFR-LEGAL-03).

4.3 files

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
user_id	UUID	FK → users.id, NULL if guest upload, ON DELETE CASCADE
guest_session_id	UUID	FK → guest_sessions.id, NULL if registered-user upload, ON DELETE CASCADE
original_filename	VARCHAR(255)	NOT NULL
file_type	VARCHAR(10)	NOT NULL, CHECK IN ('audio','video')
mime_type	VARCHAR(100)	NOT NULL
file_size_bytes	BIGINT	NOT NULL, CHECK (file_size_bytes > 0)
storage_path	VARCHAR(500)	NOT NULL
status	VARCHAR(30)	NOT NULL, DEFAULT 'UPLOADED', CHECK IN ('UPLOADED','QUEUED','PROCESSING','READY_FOR_REVIEW','REGENERATING','COMPLETED')
duration_seconds	NUMERIC(10,2)	NULL (populated after audio extraction)
detected_language	VARCHAR(20)	NULL, CHECK IN ('english','hindi','hinglish','mixed')
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()
deleted_at	TIMESTAMPTZ	NULL (soft delete)

Constraint: `CHECK (user_id IS NOT NULL OR guest_session_id IS NOT NULL)` — a file must belong to either a registered user or a guest session, never neither.

4.4 detection_configs

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
file_id	UUID	FK → files.id, NOT NULL, ON DELETE CASCADE
mode	VARCHAR(20)	NOT NULL, CHECK IN ('automatic','custom','combined')
preset_id	UUID	FK → presets.id, NULL
custom_words	JSONB	NULL (array of strings; used when mode is 'custom' or 'combined')
replacement_mode	VARCHAR(20)	NOT NULL, DEFAULT 'beep', CHECK IN ('beep','mute','custom_sound')
custom_sound_path	VARCHAR(500)	NULL (required if replacement_mode = 'custom_sound')
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

4.5 presets

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
user_id	UUID	FK → users.id, NOT NULL, ON DELETE CASCADE
name	VARCHAR(100)	NOT NULL
word_list	JSONB	NOT NULL (array of strings)
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

Constraint: `UNIQUE (user_id, name)` — a user cannot have two presets with the same name.

4.6 jobs

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
file_id	UUID	FK → files.id, NOT NULL, ON DELETE CASCADE
job_type	VARCHAR(30)	NOT NULL, CHECK IN ('initial_processing','regeneration')
status	VARCHAR(20)	NOT NULL, DEFAULT 'PENDING', CHECK IN ('PENDING','RUNNING','COMPLETED','FAILED','RETRYING')
retry_count	INT	NOT NULL, DEFAULT 0
max_retries	INT	NOT NULL, DEFAULT 3
error_message	TEXT	NULL
started_at	TIMESTAMPTZ	NULL
completed_at	TIMESTAMPTZ	NULL
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

4.7 job_logs

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
job_id	UUID	FK → jobs.id, NOT NULL, ON DELETE CASCADE
stage	VARCHAR(50)	NOT NULL (e.g., 'audio_extraction','asr','detection','muxing')
level	VARCHAR(10)	NOT NULL, CHECK IN ('INFO','WARN','ERROR')
message	TEXT	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

4.8 **detections**

Column	Type	Constraints
id	UUID	PK, default <code>(gen_random_uuid())</code>
file_id	UUID	FK → files.id, NOT NULL, ON DELETE CASCADE
detected_text	TEXT	NOT NULL
language	VARCHAR(20)	NULL
start_time	NUMERIC(10,3)	NOT NULL (seconds, 3 decimal precision for ms accuracy)
end_time	NUMERIC(10,3)	NOT NULL
confidence_score	NUMERIC(4,3)	NOT NULL, CHECK (confidence_score BETWEEN 0 AND 1)
detection_source	VARCHAR(20)	NOT NULL, CHECK IN ('dictionary','custom_list','model_inferred','regex')
dictionary_id	UUID	FK → profanity_dictionary.id, NULL
replacement_mode	VARCHAR(20)	NOT NULL, DEFAULT 'beep', CHECK IN ('beep','mute','custom_sound')
is_user_added	BOOLEAN	NOT NULL, DEFAULT false
is_dismissed	BOOLEAN	NOT NULL, DEFAULT false
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

Constraint: `CHECK (end_time > start_time)`.

4.9 **profanity_dictionary**

Column	Type	Constraints
id	UUID	PK, default <code>(gen_random_uuid())</code>
term	VARCHAR(100)	NOT NULL
normalized_term	VARCHAR(100)	NOT NULL (lowercased, normalized form used for matching)
language	VARCHAR(20)	NOT NULL, CHECK IN ('english','hindi','hinglish','universal')
severity	VARCHAR(10)	NOT NULL, DEFAULT 'medium', CHECK IN ('low','medium','high')
is_active	BOOLEAN	NOT NULL, DEFAULT true
added_by_admin_id	UUID	FK → users.id, NULL
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

Constraint: `UNIQUE (normalized_term, language)`.

4.10 `exports`

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
file_id	UUID	FK → files.id, NOT NULL, ON DELETE CASCADE
export_type	VARCHAR(20)	NOT NULL, CHECK IN ('video','audio','srt','txt','json')
storage_path	VARCHAR(500)	NOT NULL
file_size_bytes	BIGINT	NULL
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

4.11 `credits_ledger`

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
user_id	UUID	FK → users.id, NOT NULL, ON DELETE CASCADE
amount	INT	NOT NULL (positive = credit added, negative = credit consumed)
transaction_type	VARCHAR(30)	NOT NULL, CHECK IN ('signup_bonus','file_processing','manual_adjustment','plan_purchase')
reference	VARCHAR(255)	NULL (e.g., related file_id or job_id as text)
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

Notes: A user's current balance is computed as `SUM(amount)` rather than stored redundantly, avoiding sync bugs; a materialized view or cached aggregate can be added later for performance if needed.

4.12 `admin_logs`

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
admin_id	UUID	FK → users.id, NOT NULL
action_type	VARCHAR(50)	NOT NULL (e.g., 'dictionary_add','user_disable','dictionary_edit')
target_entity	VARCHAR(50)	NOT NULL (e.g., 'users','profanity_dictionary')
target_id	UUID	NULL
details	JSONB	NULL (before/after values or additional context)
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

4.13 **feedback**

Column	Type	Constraints
id	UUID	PK, default <code>gen_random_uuid()</code>
user_id	UUID	FK → users.id, NULL (feedback can be anonymous/guest)
category	VARCHAR(30)	NOT NULL, CHECK IN ('bug','feature_request','detection_accuracy','other')
message	TEXT	NOT NULL
status	VARCHAR(20)	NOT NULL, DEFAULT 'open', CHECK IN ('open','reviewed','resolved')
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()

5. Relationships Summary

Relationship	Type	ON DELETE Behavior
users → files	1-to-many	CASCADE (deleting a user removes their files)
guest_sessions → files	1-to-many	CASCADE
files → jobs	1-to-many	CASCADE
files → detections	1-to-many	CASCADE
files → exports	1-to-many	CASCADE
files → detection_configs	1-to-1 (per processing run)	CASCADE
jobs → job_logs	1-to-many	CASCADE
users → presets	1-to-many	CASCADE
presets → detection_configs	1-to-many (referenced, optional)	SET NULL
profanity_dictionary → detections	1-to-many (optional match reference)	SET NULL
users → credits_ledger	1-to-many	CASCADE
users → admin_logs	1-to-many (as admin_id)	RESTRICT (preserve audit trail even if admin account is later modified)
users → feedback	1-to-many	SET NULL (preserve feedback even if user deletes account)

6. Indexing Strategy

Table	Index	Purpose
users	UNIQUE index on <code>email</code>	Fast login lookup; enforces uniqueness
users	Index on <code>role</code>	Fast admin-role filtering
files	Index on <code>user_id</code>	Dashboard "recent files" query (FR-DASH-01)
files	Index on <code>guest_session_id</code>	Guest cleanup job lookups
files	Index on <code>status</code>	Processing queue status queries (FR-DASH-05)
files	Composite index on <code>(user_id, created_at DESC)</code>	Efficient paginated history with sort (FR-DASH-02)
jobs	Index on <code>file_id</code>	Fetch job status per file
jobs	Index on <code>status</code>	Worker polling / admin queue monitoring
detections	Index on <code>file_id</code>	Timeline Editor load (FR-TLE-01) — most frequent query in the app
detections	Composite index on <code>(file_id, start_time)</code>	Ordered transcript rendering
profanity_dictionary	UNIQUE composite index on <code>(normalized_term, language)</code>	Fast matching lookup during detection; enforces no duplicates
profanity_dictionary	Index on <code>is_active</code>	Excludes disabled terms from matching queries
presets	Composite UNIQUE index on <code>(user_id, name)</code>	Enforce uniqueness, fast preset lookup per user
credits_ledger	Index on <code>user_id</code>	Balance calculation queries
admin_logs	Index on <code>admin_id</code>	Audit trail lookup per admin
admin_logs	Composite index on <code>(target_entity, target_id)</code>	"History of changes to this entity" queries
guest_sessions	Index on <code>expires_at</code>	Efficient cleanup job querying for expired sessions

7. Data Integrity & Constraints Summary

- **Referential integrity** enforced via foreign keys with explicit cascade rules (Section 5) — no orphaned `detections`, `jobs`, or `exports` rows can exist without a parent `files` row.
- **Check constraints** enforce valid enum-like values at the database level (e.g., `status`, `role`, `detection_source`) rather than relying solely on application-layer validation — defense in depth per SRS-NFR-SEC-03.
- **NOT NULL constraints** on all fields required for core business logic (e.g., `start_time`/`end_time` on `detections`) prevent incomplete records from corrupting

the Timeline Editor UI.

- **Row-Level Security (RLS)** policies (Supabase-native) applied on `files`, `detectors`, `presets`, `credits_ledger`, and `feedback` tables to ensure a user can only query/modify rows where `user_id` matches their authenticated JWT `sub` claim — this is the primary authorization enforcement mechanism at the data layer, complementing service-layer RBAC checks (per System Architecture Section 7.1).
 - `admin_logs` is **append-only** at the application layer (no UPDATE/DELETE exposed via API) to preserve audit trail integrity.
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8. Migration & Versioning Strategy

- Schema changes managed via a migration tool (e.g., Alembic, given the FastAPI/Python backend) with version-controlled migration files committed to the repository.
 - Each migration is additive-first where possible (add nullable columns, backfill, then tighten constraints in a follow-up migration) to avoid downtime on the free-tier hosted database.
 - `profanity_dictionary` seed data (initial curated word list) is loaded via a dedicated seed script, separate from schema migrations, so it can be updated independently by admins without a deployment.
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9. Next Document

Document 7: Folder Structure — the complete production-level project directory structure (frontend, backend, AI, database, configs, tests, docs) implementing the architecture and data model defined above.

End of Database Design Document